Electronic Schematic Symbol Classifier

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Topic of Interest

We are investigating electronic schematics. Our goal is that given a handwritten electronic schematic, we can develop a model that classifies individual components in the schematic.

Summary: 1. Why Is This Topic Important?

- Hand-drawn circuit schematics are common in electronics education and early-stage prototyping, but their legibility varies widely, forcing engineers and students to manually transcribe sketches into digital formats before they can simulate or verify their designs.
- This process is slow, error-prone, and slows down iteration, especially in classrooms or collaborative settings where schematics are often drawn informally.
- By applying computer vision and object detection to automatically identify circuit components from hand-drawn sketches, this project addresses a practical bottleneck while also tackling a broader challenge: building reliable models for messy, real-world visual data, an area where well-labeled datasets are typically scarce.

Summary:

2. How Will Data Be Used?

We will train a CNN-based object detection model using a dataset of labeled, hand-drawn circuit schematic images. This data allows the model to learn the distinct visual patterns associated with each component symbol, effectively adapting to wide variations in individual handwriting and drawing styles.

3. Potential Impact

This solution will enable students and engineers to quickly verify and digitize informally sketched circuit designs without the need to manually redraw them in CAD software. On a broader scale, it proves how computer vision can be successfully applied to noisy, real-world visual data, laying a foundation for structuring other technical diagrams like flowcharts or architectural sketches.

Machine Learning Algorithms

Algorithm 1: Computer Vision (CV)

Algorithm 2: Convolutional Neural Networks (CNNs) focusing on Object Detection.

Algorithm 3: YOLO (You Only Look Once) for Object Detection

Research Question

Core Question: Can we classify essential electronic schematic components given a handwritten image of them?

Datasets

Primary Dataset

- **Name:** Digitize-HCD: A Dataset for Digitization of Handwritten Circuit Diagrams
- **Source:** Mendeley Data
- **Link:** https://data.mendeley.com/datasets/rngcz5wtv8/2#:~:text=A%20total%20of%201%2C277%20images,a%20resolution%20of%20600%20dpi.

Secondary Dataset

- **Name:** Handwritten Schematics - CGHD
- **Source:** Kaggle
- **Link:** https://www.kaggle.com/datasets/johannesbayer/cghd1152

Training & Testing Considerations

- The primary dataset (Digitize-HCD from Mendeley Data) consists of images of handwritten electronic circuit schematics with bounding box coordinates for specific components.
- It features 1,277 handwritten circuit diagrams sourced from 150 volunteers, ensuring diversity in factors such as handwriting styles and drawing techniques.
- Given our project timeline and scope, we are training on approximately 17 foundational electronic components.
- A secondary dataset includes more examples and components that we may use to expand our model if we have time.

Sources of Bias

- There are different standards used across the world which result in different symbols indicating the same parts across the world. The dataset might bias one of those standards, and thus not recognize some components.
- The data for the model is handwritten schematics, so there are conditions in the data that may influence the model such as lighting, drawing material, and more.

Mitigation Strategies

- Sample appropriately ensuring fair representation across regional standards, individual components, and other factors.
- Implement data augmentation and preprocessing to provide more data and examples to train on, and so the model learns to adapt to different conditions.

Citations

1. N. Ahmed, M.F. Adnan, A. Shafiullah, H.J. Parash, Md.S. Rahman, I.C. Akib, G. Sarowar, “Digitize-HCD: A dataset for digitization of handwritten circuit diagrams,” Data in Brief, vol. 59, p. 111315, Apr. 2025, doi: 10.1016/j.dib.2025.111315
2. Handwritten Schematics - CGHD [Dataset]. Kaggle. https://doi.org/10.34740/KAGGLE/DSV/13548113
3. S. Amraee, M. Chinipardaz, M. Charoosaei and M. A. Mirzaei, "Handwritten Logic Circuits Analysis Using the YOLO Network and a New Boundary Tracking Algorithm," in IEEE Access, vol. 10, pp. 76095-76104, 2022, doi: 10.1109/ACCESS.2022.3192467.